

ECONOMICS

Week 02

Time Value of Money & Interest Calculations

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Terminologies

- Adjusted (cost) basis:
The original cost basis of the assets, adjusted by allowable increase or decrease is used to compute depreciation deductions or additions
- Unadjusted cost basis:
The initial value of acquiring an asset (purchase price + any sales tax) including transportation expenses and other normal costs of making the asset serviceable for its intended use; also called unadjusted cost basis.
- Book Value (BV):
The original cost basis of the property, including any adjustments, less all allowable depreciation deductions.
- Market Value (MV):
The present value of what will be received through ownership of the property.

Terminologies

- Recovery Period:
The number of years over which the basis of the property is recovered through accounting process.
- Recovery rate:
A percentage (expressed in decimal form) for each year of the recovery period which is applied for the deduction of depreciation.
- Salvage Value:
The estimated value of a property at the end of its useful life. It is the expected selling price of a property when the asset can no longer be used.
- Useful life:
The expected (estimated) period that a property will be used in a trade or business to produce income. It is not how long the property can last but how long it can be used productively.

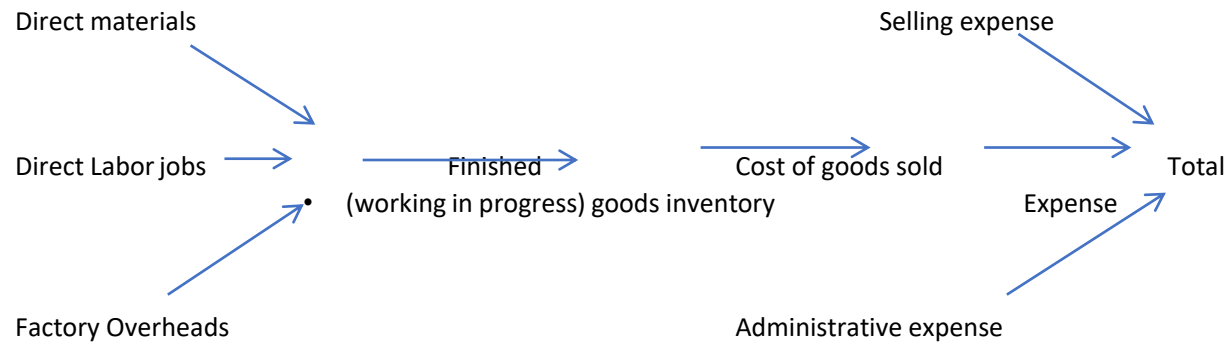
Interest Terminologies

- Interest:
 - It is the fee which is charged in using someone else's money
- Interest Rate:
 - The percentage of money which is charged as Interest is called Interest Rate
- Simple Interest:
 - Fixed Percentage of the Principal (amount of money borrowed) multiplied by the life of the loan.
- Compounds Interest:
 - When Interest is compounded total time period is subdivided into several interest periods (e.g., one-month, three-month, six-month etc.). Interest is credited at the end of each Interest Period and is allowed to accumulate from one interest period to the other.

Accounting Equation

Assets	=	Liability	+	Owners Equity
Cash		Short-term debt		Capital Stock
Receivables		Payables		Retained Earnings
Inventories		Long-term debt		(income earnings retained in the firm)
Equipment				
Buildings				
Land				

Cost Accounting



Notation and Cash-Flow Diagrams and Tables

The following notation is utilized in formulas for compound interest calculations:

i = effective interest rate per interest period;

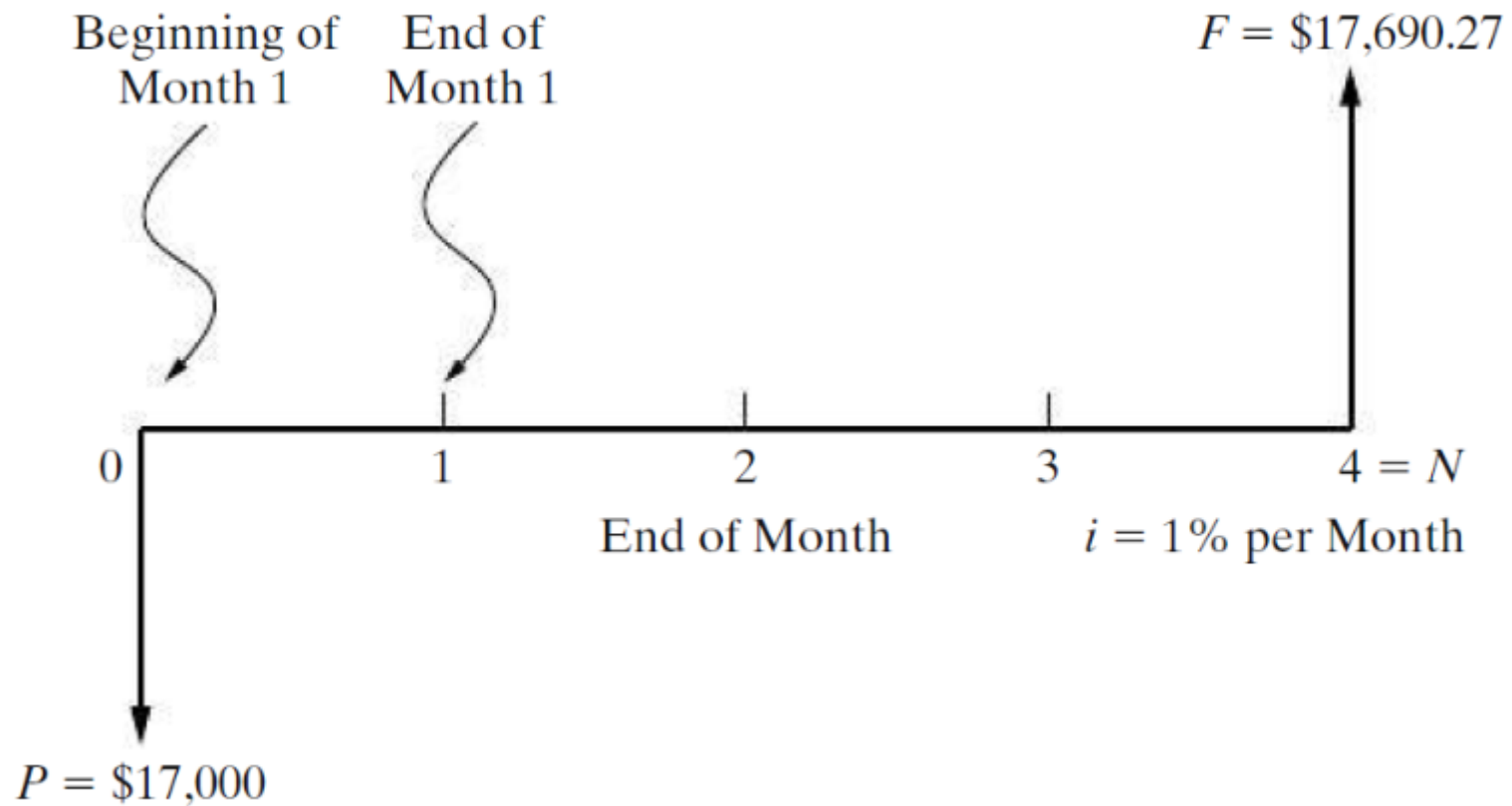
N = number of compounding (interest) periods;

P = present sum of money; the equivalent value of one or more cash flows at a reference point in time called the present;

F = future sum of money; the equivalent value of one or more cash flows at a reference point in time called the future;

A = end-of-period cash flows (or equivalent end-of-period values) in a uniform series continuing for a specified number of periods, starting at the end of the first period and continuing through the last period.

Cash-flow diagram

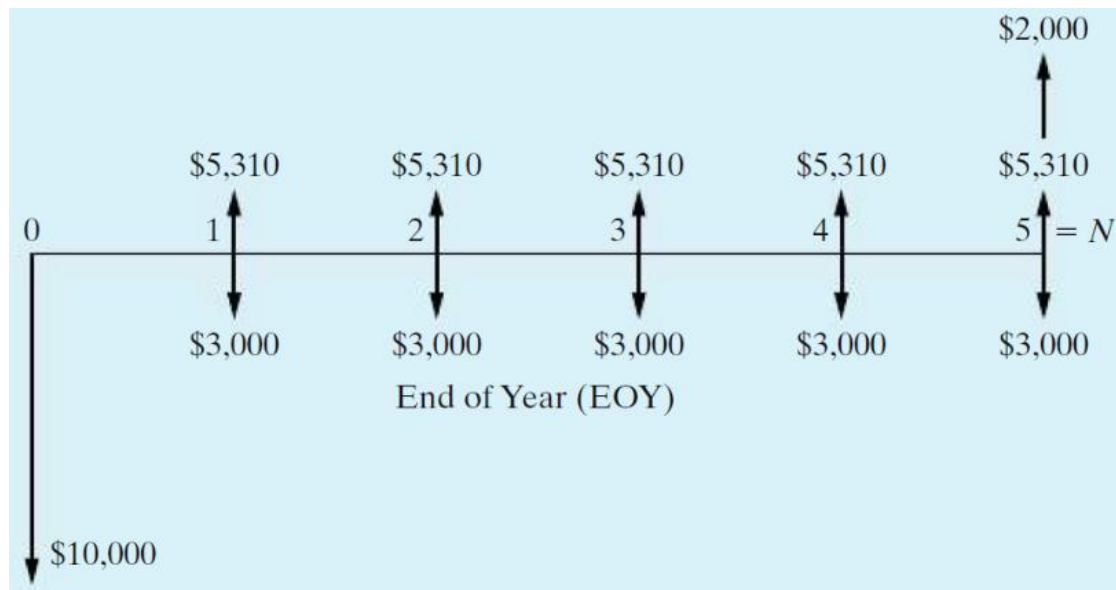


Cash-flow diagram - Example

Before evaluating the economic merits of a proposed investment, the XYZ Corporation insists that its engineers develop a cash-flow diagram of the proposal. An investment of \$10,000 can be made that will produce uniform annual revenue of \$5,310 for five years and then have a market (recovery) value of \$2,000 at the end of year (EOY) five. Annual expenses will be \$3,000 at the end of each year for operating and maintaining the project. Draw a cash-flow diagram for the five-year life of the project.

Cash-flow diagram - Solution

As shown in the figure below, the initial investment of \$10,000 and annual expenses of \$3,000 are cash outflows, while annual revenues and the market value are cash inflows.



Notice that the beginning of a given year is the end of the preceding year. For example, the beginning of year two is the end of year one.